

4.0 Java Programming Language

4.1 Introduction to Java Program

Introduction & Components of Java

COMPUTER SCIENCE II| SC025

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Learning Outcomes

4.1 Introduction to Java Program

a. Describe Object-Oriented Programming.

b. Describe the components of a Java program: comments, class, main method, body.

Object-Oriented Programming Structure

Java is an **object-oriented programming language** (**OOP**) founded in 1995 owned by Oracle.

Advantages of OOP:

- OOP is faster and easier to execute.
- OOP provides a clear structure for the programs.
- OOP makes the code easier to maintain, modify and debug by reducing redundancy in code.
- OOP makes it possible to create full reusable applications with less code and shorter development time.

The **structure** of **object-oriented programming** (OOP) covers **class**, **objects**, and **methods**.

Class

Definition:

Class is a **description** of the **properties** and the **behaviors** of one or more **objects**.

- Think of it as a generic blueprint or template that defines the structure & behaviour of objects.
- In Java, the *class* keyword is used to create a class.
- The fields of a class are to define the properties of its objects.
- The methods of a class are to define the behaviors of its objects (what they can do…)

```
class ClassName{
```

Objects

Definition:
Object is an **entity** that has both **state** and **behavior**.

- Objects are specific **instances created from a class**, that has access to methods and attributes (fields) defined in the class.
- Objects have properties (data values) and behaviors (methods) that manipulate the data.
- When an object is created, memory is allocated and specific values can be stored in its attributes.
- Object-oriented programs use objects as they are the building blocks of OOP.

Methods

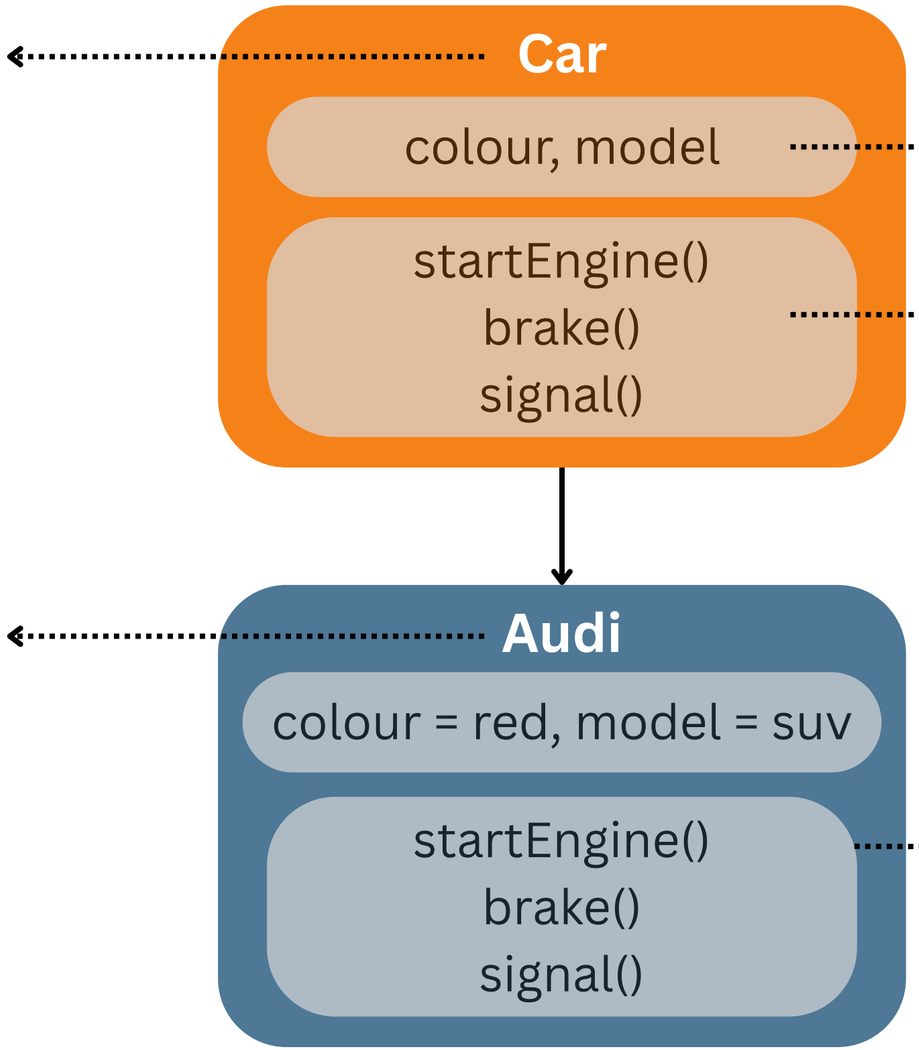
Definition:

Method is a **collection of statements** that perform **specific tasks** and **returns the result** to the caller.

- Provides easy modification and readability to code by adding / removing just a chunk of the code.
- Good for code reusability as we only need to write the method once then we can use is as many times as we want.
- Methods are only executed (run) when a method call is made in the *main* method.

Class, Objects & Methods Example

Car class that has the attributes *colour* & *model* and the methods *startEngine(), brake(), and signal();*



Attributes (variables)

Methods (functions)

An instance of the Car class called *Audi*. *Audi* is an object created from the Car class.

Objects inherit attributes and methods from their class.

Class, Objects & Methods Example

In a game of Mobile Legends, there are many classes of heroes. This can also be used as an example to show how class, objects, and methods relate.

Class: Mage



Properties (Fields)	Methods
Name	attack( )
HP	levelUp( )
Mana	skill1( )
Physical Attack	skill2( )
Level	ultimate( )

Object: Valir



Properties (Fields)	Methods
Name: Valir	attack( )
HP: 2390	levelUp( )
Mana: 500	skill1( )
Physical Attack: 105	skill2( )
Level: 1	ultimate( )

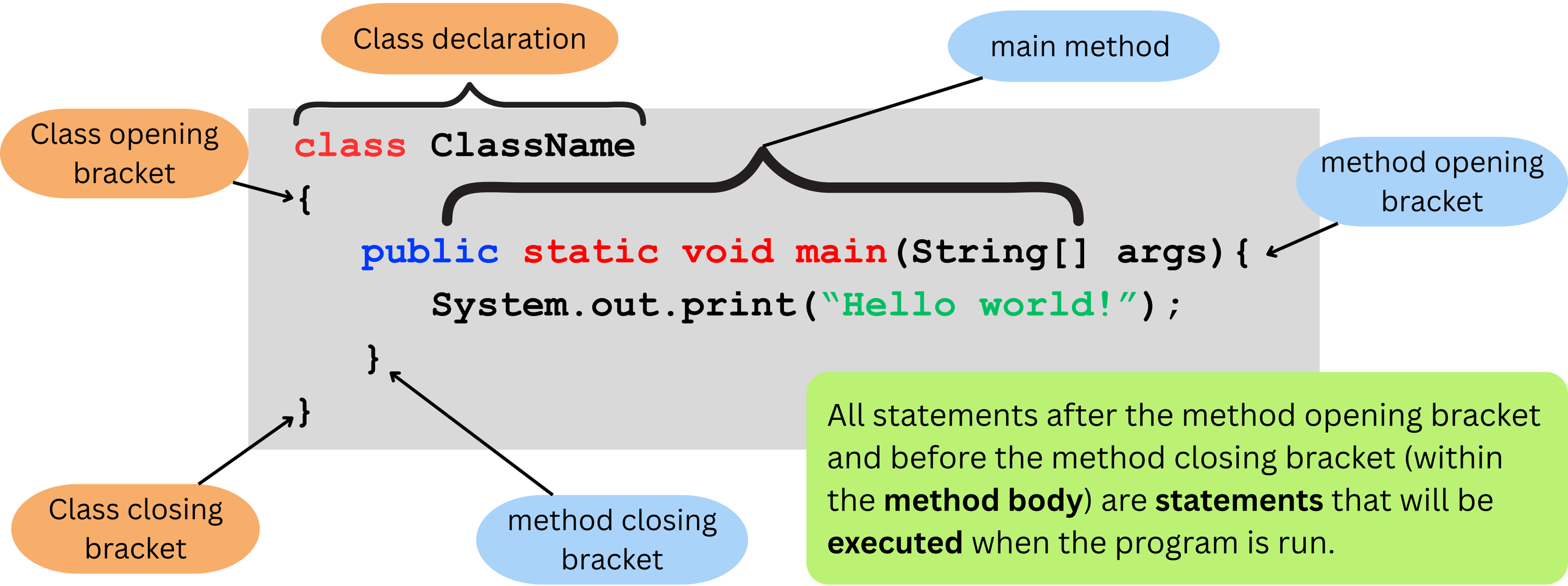
Main method

- Every Java program starts its **execution** from the ***main*** method.

```
class ClassName
{
    public static void main(String[] args){
        System.out.print("Hello world!");
    }
}
```

- The *main* method goes inside the class and is where operations will take place.
- In JAVA, we use curly braces { } to denote containers and indicate which parts of code belong where.

Main method



All statements after the method opening bracket and before the method closing bracket (within the **method body**) are **statements** that will be **executed** when the program is run.

Comments

Comments are **text** written **alongside code** that will **not be executed** when the **program** is **run**.

Importance of writing comments in code:

- Makes code more readable by adding details of code.
- Easier to maintain code in the future and find errors easily
- Prevent execution of alternative code segments during testing
- Provide additional information about the program, variable, method, class, operation, or any statement.

There are **three** (3) types of comments in Java:

- **Single-line** comments
- **Multi-line** comments
- **Documentation** comments

Single-line Comments

- Used to comment **one line** of code.
- Starts with two forward-slash (//).

```
double area = length * width; //calculate area of rectangle
```

Multi-line Comments

- Used to comment **multiple lines** of code.
- Used for explanation of complex codes or to comment multiple lines.
- Comments are placed between /* and */

```
/* This code switches the
values of two variables */
```

```
/*
int age = 0;
double height = 0;
*/
```

Java Documentation Comment

Documentation comments are used to generate external documentation using Javadoc.

- Usually used to **describe** classes, methods, and parameters.
- The comments are places between /** and */

```
/**
 * This program calculates the average of three numbers entered
 * by user.
 * author John Wick
 * version 1.0
 */
```